SC Honors Geometry - Nappier
Homework 10.7 – Circles in the Coordinate Plane

Name: ____________________________________

Write the equation.

1. Center (0, 0), radius 6.

2. Center (2, −5), radius 4.

3. Identify the center and radius of $(x + 3)^2 + (y - 1)^2 = 49$.

Radius from a point.

4. Center (0, 0), passing through (−6, 8). Find r and write the equation.

5. Center (4, −3), passing through (1, 1). Find r and write the equation.

6. A diameter has endpoints (−2, 3) and (6, −3). Find the center, radius, and equation.

On the circle?

7. Is (5, −2) on the circle $(x - 2)^2 + (y + 2)^2 = 9$?

8. Is (−3, 4) on the circle $x^2 + y^2 = 25$? Is (3, 3) inside or outside?

From a graph.

9. A graphed circle has center (3, −4) and radius 3. Write its equation.

10. Sketch $x^2 + y^2 = 16$ on your own grid paper. Name all four points where it crosses the axes.